

Deployment Guide

JSPL EAF Industrial AI Decision Support — Release 1.0
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Prerequisites

- Python 3.11+, Node 20+
- CatBoost, pandas, scikit-learn (backend requirements)
- Research folders `phase\_19` through `phase\_32` present at repo root

Backend

```
```bash
cd backend
pip install -r requirements.txt
uvicorn app.main:app --host 0.0.0.0 --port 8001
```
```

Verify: `GET http://localhost:8001/health` → version 3.0.0

Frontend

```
```bash
cd steelops-ai/frontend_v2
npm install
npm run build
npm run dev
```
```

Set `NEXT\_PUBLIC\_EAF\_API\_URL=http://localhost:8001`

Phase 33 data directories

- `backend/data/phase\_33/validation\_results.json`
- `backend/data/phase\_33/operator\_feedback.json`

Research deliverables

```
```bash
python research/phase_33_industrial_product_validation/phase_33_pipeline.py
```
```

Readiness checklist

- [] Prediction default $120\text{ t} \approx 39.9\text{ min}$
- [] `/hybrid/evaluate` returns reliability 68–78 on live heats
- [] `/optimize` unchanged from Phase 20.2
- [] `/optimize/v2` never changes POWER
- [] Validation page records heats
- [] Deployment readiness page loads traffic lights